

Pix2Repair: Implicit Shape Restoration from Images

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Abstract

We present Pix2Repair, an automated shape repair approach that generates restoration shapes from images to repair fractured objects. Prior repair approaches require a high-resolution watertight 3D mesh of the fractured object as input. Such input 3D meshes must be obtained using expensive 3D scanners and require manual cleanup, limiting accessibility and scalability. Pix2Repair takes an image of the fractured object as input and automatically generates a 3D printable restoration. We contribute a novel shape function that deconstructs a latent code representing the fractured object into a complete shape and a break surface. We collect a dataset of fractured everyday objects and show restorations for real fractures from our dataset and for synthetic fractures from the Geometric Breaks and Breaking Bad datasets. Our approach outperforms shape completion approaches adapted for shape repair in terms of chamfer distance, normal consistency, and percent restorations generated.

1 Introduction

We all know the frustration of trying to repair a treasured glass, vase, or mug that has been accidentally fractured, e.g., Figure 1(a). Sometimes fractured objects can be repaired via reassembly. However, in many cases parts are destroyed or lost during the fracturing process leaving gaps in the repaired object. Automated shape repair approaches seek to fill this gap by automatically generating restoration shapes for fractured objects. In addition to preserving heirlooms, object repair algorithms have the potential to reduce household waste, restore cultural heritage objects, and automate industrial recycling. A recent body of work has emerged to automate the repair of fractured objects by predicting 3D printable restoration shapes using machine learning [Hermoza and Sipiran, 2018, Lamb et al., 2022c, Lamb et al., 2022b, Lamb et al., 2022a, Jaramillo and Sipiran, 2024]. However, existing approaches operate in low-resolution voxel space [Hermoza and Sipiran, 2018], or require high resolution 3D scans of the fractured object [Lamb et al., 2022c, Lamb et al., 2022b, Lamb et al., 2022a, Jaramillo and Sipiran, 2024] which must be obtained using expensive scanners, e.g., the \$2,000 (USD) Einscan-SP, making them inaccessible to consumers.

We present Pix2Repair, an approach to automatically generate repair parts for fractured objects from images. Our approach builds on prior approaches that perform shape completion from images using implicit representations [Murez et al., 2020, Lin et al., 2020, Mescheder et al., 2019, Wang et al., 2022, Xu et al., 2020, Chen and Zhang, 2019, Li and Zhang, 2021, Hui et al., 2022, Cheng et al., 2023]. However, existing shape completion approaches generate a complete shape from a partial observation, e.g., an image that observes the complete shape from a fixed perspective. Unlike traditional shape completion, our approach generates a restoration shape from an image of a fractured object, such as the image shown in Figure 1(a). Generating restoration shapes is a harder task than generating complete shapes as restorations are often much smaller than complete shapes, as is the case for the restoration shape in Figure 1(f). Restorations also contain high-frequency geometry at the point of fracture that is difficult to predict directly.

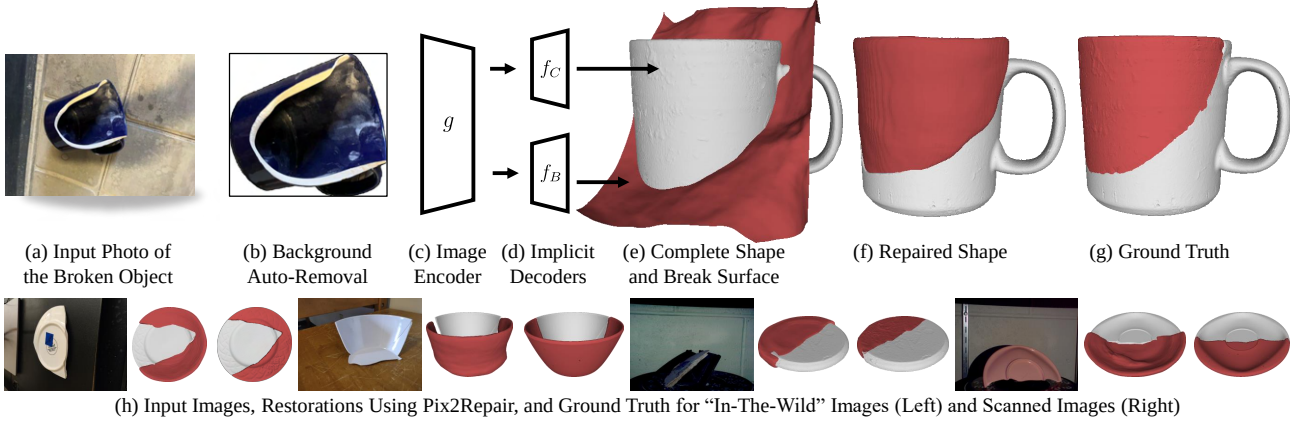


Figure 1: We present Pix2Repair, an automated shape repair approach. Pix2Repair takes an image of a fractured object as input, automatically removes the background, and predicts a complete shape and break surface, which combine to generate a restoration shape.

To overcome the challenge of representing shapes with small size and high-frequency surface detail, we compute the repair shape as the Boolean intersection of a predicted complete shape and break shape, where the break shape is given by a break surface, shown in Figures 1(e). Our approach draws inspiration from existing shape repair approaches [Lamb et al., 2022b, Lamb et al., 2022a], which similarly represent the restoration shape as a Boolean intersection. However, our approach learns to predict a complete shape and break surface from a single shape code representing the fractured object, in contrast to prior approaches that obtain two separate complete and break codes by performing iterative optimization during inference. Learning a mapping from the fractured shape code to the complete shape and break surface allows our network to predict restoration shapes directly from color images of the fractured shape. Accepting images as input makes Pix2Repair more accessible to consumers and faster during inference than prior approaches.

2 Related Work

Approaches to perform repair of damaged objects use voxels [Hermoza and Sipiran, 2018], point clouds [Jaramillo and Sipiran, 2024], or leverage implicit representations [Lamb et al., 2022b, Lamb et al., 2022c, Lamb et al., 2022a]. Approaches based on implicit representations learn a mapping from fractured shapes to restoration shapes [Lamb et al., 2022c], or learn occupancy [Lamb et al., 2022b], or combined signed distance, occupancy, and normal field functions [Lamb et al., 2022a] for a complete shape and break surface that combine to generate the restoration shape. Unlike these approaches that use 3D shape as input, our work uses images, enabling repair from a single image. Work exists in generative prediction of brittle fractures [Huang and Kanai, 2025] and in understanding images of cultural heritage objects using vision language models [Zhang et al., 2026]. Approaches have also investigated re-assembly [Chen et al., 2022, Tsesmelis et al., 2024], where the goal is assembling existing pieces rather than inferring new missing ones. A related field of study is shape completion, which focuses on generating a complete shape from single and multi-view images [Xie et al., 2020, Wu et al., 2023, Yi et al., 2021, Liu et al., 2023, Chen et al., 2026, Wang et al., 2021, Cheng et al., 2023]. A large body of recent work has focused on learning to represent shapes implicitly using functions [Murez et al., 2020, Lin et al., 2020, Chen and Zhang, 2019, Wang et al., 2022, Li and Zhang, 2021, Xu et al., 2020, Hui et al., 2022, Mescheder et al., 2019, Wang et al., 2021], including methods to learn multi-scale implicit fields [Wang et al., 2022], disentangle object geometry and texture [Li and Zhang, 2021], impose symmetry constraints [Xu et al., 2020], or deforming shape templates [Hui et al., 2022]. Unlike shape completion approaches, our approach learns to predict a restoration shape as the Boolean intersection of two

other shape functions, i.e., the complete shape and break surface.

3 Pix2Repair

We use the approach from DeepMend [Lamb et al., 2022b] to represent a restoration shape R as arithmetic relaxations to logical operations between learnable occupancy function representations o_C and o_B of a complete shape C and a break surface B , as $o_R(\mathbf{x}) = o_C(\mathbf{x})o_B(\mathbf{x})$, where $\mathbf{x} \in \mathbb{R}^3$ represents a 3D point. We learn two functions, f_C and f_B , to represent occupancy in the complete shape and occupancy in the break shape respectively. During training, the function f_C is tasked with learning a one-to-many mapping from the fractured shape observation to the complete shape. The function f_B is tasked with learning a one-to-one mapping from the fractured shape observation to the break shape. Each function takes point $\mathbf{x} \in \mathbb{R}^3$ as input and is conditioned on a latent code corresponding to an observation of the fractured shape $\mathbf{z}$, such that $o_C(\mathbf{x}) = f_C(\mathbf{z}, \mathbf{x})$ and $o_B(\mathbf{x}) = f_B(\mathbf{z}, \mathbf{x})$. During training and testing we obtain $\mathbf{z}$ by passing an input image through an image encoder g to obtain a fractured shape latent code. We compute occupancy in the restoration shape by substituting the predicted values into its occupancy representation, i.e., as $o_R(\mathbf{x}) = f_B(\mathbf{z}, \mathbf{x})f_C(\mathbf{z}, \mathbf{x})$.

We train our network on complete, break, and restoration shape tuples (C, B, R) . We use $\hat{o}_C(\mathbf{x})$, $\hat{o}_B(\mathbf{x})$, and $\hat{o}_R(\mathbf{x})$ to denote the ground truth occupancy values for shapes C , B , and R respectively at a point $\mathbf{x}$. During training, for each shape tuple we optimize with respect to the loss

$$\mathcal{L} = \sum_{\mathbf{x} \in \mathcal{X}} (BCE(o_C(\mathbf{x}), \hat{o}_C(\mathbf{x})) + \lambda_B BCE(o_B(\mathbf{x}), \hat{o}_B(\mathbf{x})) + \lambda_R BCE(o_R(\mathbf{x}), \hat{o}_R(\mathbf{x}))). \quad (1)$$

In Equation (1), BCE is the binary cross-entropy loss, and $\mathcal{X}$ is a set of probing points. The individual loss terms encourage the predicted values for the complete, break, and restoration shapes to converge to the ground truth values during training. λ_B and λ_R are coefficients that allow us to balance the relative importance of the complete, break, and restoration shape loss. During training we optimize with respect to the parameters of g , f_C , and f_B . We use $\lambda_B = 1.0$ and $\lambda_R = 1.0$. To perform optimization we use the Adam optimizer with a learning rate of $lr = 2e^{-5}$. For all experiments we train our approach for 10 epochs. Unless otherwise specified we use ResNet34 as our image encoder g . We concatenate ResNet34 with a fully connected layer with 512 output nodes to obtain a 512 dimensional fractured shape code $\mathbf{z}$. We represent the occupancy functions f_C and f_B using two neural networks with 6 ResNet blocks and batch normalization. Each ResNet block has 256 nodes with leaky ReLU activation functions. The latent code and points are input to f_C and f_B separately using two fully connected layers with 256 output nodes and added together after the first layer. The final layer has a single output node with a sigmoid activation function.

To obtain sample points for training, for each training shape tuple we pre-compute n sample points uniformly in a cube with side lengths of 1.1 units surrounding the shapes. We also pre-compute n sample points on the surface of each shape in the tuple, and offset the surface points randomly according to a Gaussian distribution with $\sigma = 0.01$. We combine the uniform and surface sampled points into a single set $\mathcal{P}$. We use $n = 50,000$. On each training iteration we randomly select a subset $\mathcal{X} \subset \mathcal{P}$ of m sample points from the pre-computed uniform and surface sampled points. We select the points $\mathcal{X}$ such that, for each shape $S \in \{C, B, R\}$, at least $\frac{m}{6}$ points lie inside of S and at least $\frac{m}{6}$ points lie outside of S . During training, we use $m = 2,048$ sample points. After training our network, to obtain the restoration mesh we query our network with a grid of k^3 points and extract the 0.5 level isosurface using marching cubes. For evaluation we use $k = 128$. For the visualizations shown in the paper we use $k = 256$.

We input images of fractured objects with backgrounds removed to our network. We separately train a background auto-remover to automatically detect the fractured object and remove the background using the images and ground truth masks from our dataset. We use Segment Anything Model (SAM) as the base model and fine-tune its mask decoder to generate favorable masks based on the approach proposed by MedSAM [Ma et al., 2024]. A bounding box of the same size as the input image is used as the prompt for the decoder. To further remove the noise, only the largest connected component in the generated mask is returned as the final mask. We use Dice Similarity Coefficient (DSC) to evaluate the removal performance. Our auto-remover achieves an average DSC score of 0.9608, while the original SAM achieves a score of 0.0001.

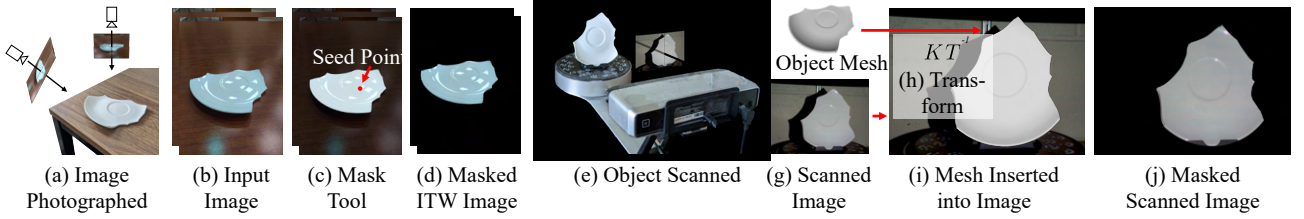


Figure 2: Data collection. For ITW images, we (a-b) photograph objects, and (c) use SAM to (d) obtain masked images. For scanned images, we use the model and .xml files obtained from Einscan scanner 3D scan process in *Fantastic Breaks* (scanning process shown in (e)), to (f-g) obtain a mesh and image, and use the transform in (h) to (i) insert the mesh into the image and (j) obtain the masked image.

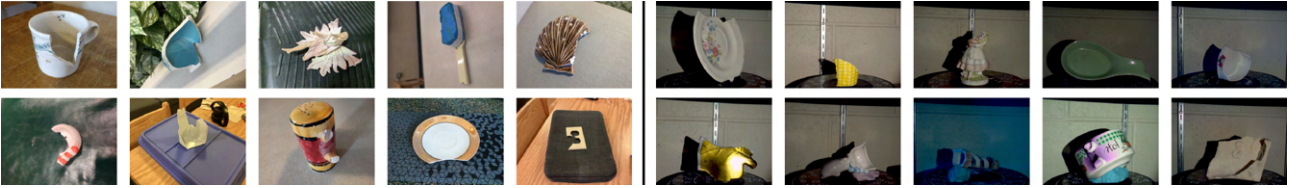


Figure 3: Example ITW photographs (left set) and scanned images (right set) from our dataset.

4 Experiments

We train Pix2Repair on different combinations of two synthetic datasets, as well as a dataset of real images that we collected, and evaluate it on our in-house dataset. Since no image-based shape repair approaches exist, for comparison, we adapt two existing shape completion approaches, Pix2Vox++ [Xie et al., 2020] and Occupancy Networks (ONet) [Mescheder et al., 2019], to take images as input to perform shape repairs.

In-house Dataset of Real Images. We create an in-house dataset of 11,637 real-world images of damaged objects from *Fantastic Breaks* [Lamb et al., 2023], allowing shape repair approaches to use 3D models from *Fantastic Breaks* for training and evaluation. 9,528 images are taken “in-the-wild”, i.e., photographed in a variety of residential, office, and classroom environments, and 2,109 scanned images from the imaging process used to generate 3D scans for *Fantastic Break*. As shown in the left set of Figure 3, the 9,528 in-the-wild images are obtained using consumer grade phones and cameras. We recruited 14 experimenters to obtain photographs. Each object was photographed by at least two experimenters, from multiple viewpoints in at least three locations, as shown in Figure 2(a). Photographs were taken in diverse locations and against a variety of backgrounds within university premises. Two experimenters manually selected a seed point as input to SAM, and selected a segmentation level (out of three possible levels) that best segmented the object, as shown in Figure 2(c). Figure 2(d) shows a masked image example. We obtain the 2,109 scan images by parsing scan files of the objects provided by *Fantastic Breaks* through scanning the object using the Einscan-SP 3D scanner. The scan process is shown in Figure 2(e). We collected images from the scan files obtained by the authors of *Fantastic Breaks*. During scanning, the scanner software produces a single .xml project file that contains pointers to the color images and depth scans, as well as the scanner intrinsic matrix. Color images are saved as .bmp files. Depth scans are saved as binary .rge files, which also store the camera frame, distance from the camera to the scanner turntable, depth scan points, colors, and normals, and the transformation matrix T that aligns the depth scan to the final high-resolution 3D model. We implement a parser for the .rge files that enables extraction of the depth scan points, colors, and normals, and the object pose in the image T . We used our parser to extract colored point clouds with normals that correspond to the depth scans obtained by the scanner. We show example scanned images at the bottom of Figure 3. We obtained the mask of the object in each .bmp image using the transformation matrix T stored in the .rge file. We invert T to obtain a transform that inserts the 3D model in the .bmp image, i.e., T^{-1} , apply T^{-1} to the 3D model, and project the transformed model onto the image using the camera intrinsic matrix K to obtain the image mask, as shown in Figure 2(i).

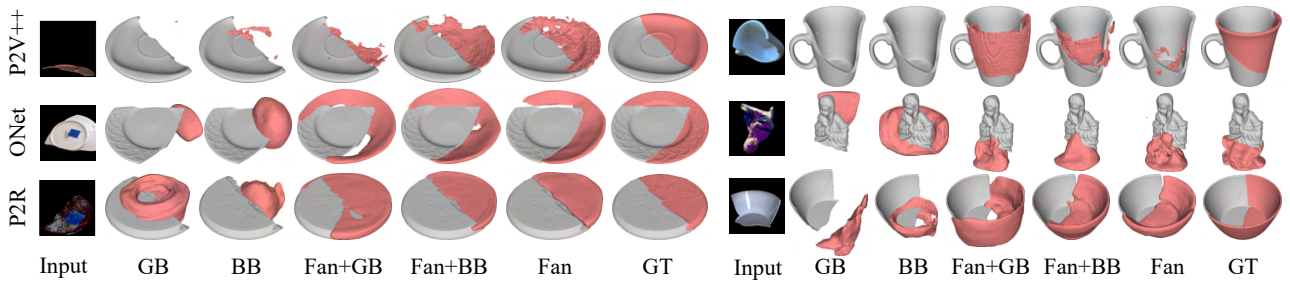


Figure 4: Repairs for images from our dataset for Pix2Vox++ (P2V++), ONet, and Pix2Repair (P2R) over various train sets and ground truth (GT). Fractured object in grey, repair in red.

Synthetic Datasets. We use two publicly available fractured object datasets that contain synthetically fractured objects: Geometric Breaks (GB) [Lamb et al., 2022a] and Breaking Bad (BB) [Sellán et al., 2022]. GB contains 25,250 objects, and each object is fractured between 1 and 10 times. Objects are fractured by subtracting a randomized geometric primitive from each object. We use objects from the jars, bottles, mugs, and GSO subsets of the dataset, totaling 5,252 objects. BB contains 10,474 publicly available 3D models. The dataset is divided into everyday objects, artifact objects, and other objects based on application. Fractures are synthesized by computing a set of the most likely fracture modes. Each object is fractured 80 times. We use 405 objects from the everyday subset, totaling 32,400 objects. All objects from the GB and BB datasets are normalized to sit inside a unit cube. Though all objects from GB are oriented upright, some objects from BB are not. We manually find and remove objects from BB that are oriented perpendicular to canonical orientations from ShapeNet. To enable computing the occupancy, we also discard BB objects that are not watertight. After removal, we have 31,661 BB objects. We generate renders that look directly at the fractured region. We follow prior work that trains on synthetic renders [Hui et al., 2022, Chen and Zhang, 2019, Huang et al., 2022, Mescheder et al., 2019]. To create renders, we compute the volumetric center of all vertices on the fractured region. We place a camera looking at the center of the fractured region vertices. We fit a plane to the fractured region vertices, and offset the camera in the direction of the plane normal. The normal may be pointing in the incorrect direction, i.e., pointing into the fractured object. To prevent incorrect normals, we compute the average dot product between the normal and the set of vectors pointing from the center of the fractured region vertices to the vertices in the fractured object, and flip the normal if the dot product is positive. We offset the camera in the direction of the normal such that it observes the entire object. We light the object using ambient lighting and using a camera-centered spotlight. We render objects at 224×224 pixel resolution using Pyrender. We split each dataset into 70%/30% train/test sets within classes. As GB contains a high number of diverse objects (5,252) and a low number of fractures per object (1-10), we randomly select five renders from each fractured object in the dataset, giving us 18,170 training and 8,090 testing renders. As BB contains a low number of diverse objects (405) and a high number of fractures per object (80), we randomly select one render from each fractured object, giving us 22,219 training and 10,181 testing renders. We use the same split as *Fantastic Breaks* for our dataset, giving us 8,598 training and 3,039 testing images. We do not allow train/test object sets to overlap.

Adapted Shape Repair Approaches. We adapt ONet and Pix2Vox++ to perform shape repair using images as our baselines. ONet represents shapes implicitly using the occupancy function. Pix2Vox++ represents shapes as voxels. To adapt shape completion approaches for shape repair, we feed images of fractured objects as input and train each approach to predict the restoration shape. For Pix2Vox++ we implement their 128^3 voxel resolution network, and train the network on 128^3 resolution voxelized restoration shapes. For ONet, we use our point sampling approach to obtain 100,000 uniform and surface sampled points.

Train Set	Pix2Vox++			ONet			Pix2Repair		
	NZ % $\uparrow$	CD $\downarrow$	NC $\uparrow$	NZ % $\uparrow$	CD $\downarrow$	NC $\uparrow$	NZ % $\uparrow$	CD $\downarrow$	NC $\uparrow$
GB	8.2%	0.230	0.518	83.9%	0.165	0.520	95.5%	0.125	0.497
BB	84.9%	0.147	0.530	87.0%	0.095	0.545	99.7%	0.068	0.577
BB + GB	36.49%	0.193	0.497	87.4%	0.121	0.528	99.2%	0.092	0.548
Ours + GB	83.1%	0.103	0.532	99.4%	0.062	0.635	99.9%	0.062	0.599
Ours + BB	86.3%	0.108	0.533	99.5%	0.051	0.641	100.0%	0.051	0.646
Ours + BB + GB	76.0%	0.134	0.538	99.7%	0.056	0.642	99.9%	0.055	0.637
Ours	90.4%	0.087	0.532	99.9%	0.049	0.638	100.0%	0.055	0.617

Table 1: Non-zero percentage (NZ%), chamfer distance (CD), and normal consistency (NC) for repair approaches. GB=Geometric Breaks, BB=Breaking Bad, Ours=Our in-house dataset. Best values are in bold.

5 Results

We show the quantitative results on our dataset in Table 1 and example restorations for objects from our dataset generated by Pix2Repair in Figure 1(h) and 4. The results demonstrate that Pix2Repair has the best performance in all but one case. They also illustrate that training repair approaches on the images of real fractured objects or synthetic datasets augmented with such real images is necessary to generate accurate restorations of real objects. To evaluate predicted restoration shapes, we use the chamfer distance (CD). CD is an indicator of overall shape similarity, but may fail to capture out-of-distribution predictions [Wu et al., 2021]. A challenge of predicting canonically oriented restoration shapes is that the pose of the restoration shape cannot be determined for objects with rotational symmetry. As the location of the fractured region is independent of the object’s canonical orientation, the network cannot accurately estimate where the restoration shape should be placed relative to the canonically oriented fractured shape. To evaluate symmetric shapes, we rotate the restoration shape around the up-axis P times, and take the rotation that minimizes the CD between the predicted restoration shape and the ground truth shape to be the pose of the predicted restoration shape. We use $P = 36$ for all classes in the dataset. We also evaluate shapes using the normal consistency (NC) [Mescheder et al., 2019]. NC captures the alignment of surface normals between the predicted and ground truth shapes. We also show the non-zero percentage (NZ%), that denotes the percentage of fractured shapes for which a restoration shape is generated, i.e., for which the occupancy is not predicted as all zero. We compute CD and NC over 30,000 points.

We show the results of training on the images of synthetically fractured objects from Geometric Breaks (GB) and Breaking Bad (BB) and evaluating on our dataset in the first 3 rows of Table 1. We observe that all approaches struggle to generate accurate repair parts for objects when trained on synthetic data, as shown by the relatively high CD, e.g., of 0.125, 0.068, and 0.092, and 0.165, 0.095, and 0.121 for Pix2Repair and ONet when trained on GB, BB, and BB + GB respectively, compared to the bottom 4 rows of Table 1. Qualitative results in Figure 4 also show visually poor repairs for GB and BB. Pix2Vox++ struggles to repair a large percentage of objects, e.g., only 8.4% when trained on GB, as small objects may be reduced to as few as 100 points during voxelization. We also observe networks trained on BB outperform networks trained on GB in terms of CD and NC, as shown in rows 1 and 2 of Table 1, likely as the physics-based fracture method of Breaking Bad is more realistic than the geometric method of Geometric Breaks.

The results of training on datasets that contain both synthetic and real fractured object images are shown in rows 4-7 of Table 1. For comparison, we also show results training only on objects from our dataset. We observe that adding real images to any combination of synthetic datasets increases performance, e.g., a CD of 0.125 to 0.062 and of 0.068 to 0.051 using Pix2Repair for GB and BB. We also observe that using our dataset without any synthetic data performs comparably to training using synthetic data with real data, e.g., with a CD of 0.051 with Ours+BB and 0.055 for ours using Pix2Repair, and with a CD of 0.051 with Ours+BB and 0.049 with ours using ONet, indicating that for some objects our image dataset contains sufficient variation to generalize to new 3D shapes. Figure 4 shows that most approaches trained on real data produce visually appealing repair parts, e.g., the plate for ONet and the coaster and bowl for Pix2Repair. Similar to training on synthetic datasets only, Pix2Repair outperforms shape completion approaches adapted for completion, except

for ONet on Fan. We also observe that combining Breaking Bad with our dataset produces some of the best repairs in terms of CD and NC, as shown in row 5 of Table 1 and column 5 of Figure 4, e.g., with a CD of 0.051 for ONet and NC of 0.646 for Pix2Repair.

6 Conclusion

We present Pix2Repair, an approach to automatically generate restoration shapes from images. Our approach outperforms existing shape completion approaches adapted for shape repair. In contrast to existing approaches that require high-resolution 3D models, by operating on images of fractured objects Pix2Repair represents a significant step forward in increasing the accessibility of repair algorithms. A limitation is that a single image of the fractured object may not observe parts of the object that are essential to determining its overall structure. Future work may investigate the use of vision-language models to incorporate interactive interfaces that guide the user when capturing the initial image or image set of the fractured object. Another limitation is that restoration shapes predicted by our approach are not accurate to scale, preventing them from being used directly for 3D printing. Accurately determining object scale from color images without external information is challenging. One approach to retain scale is to use depth cameras. Future work may explore the use depth cameras found on smartphones or augmented reality glasses.

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